

EXPERIMENT NO. : 5

DATE: 10/09/2026

1. AIM

To perform and verify Elmore delay analysis for a given RC lumped model and determine the delay at different nodes for different points of interest.

2. TOOL

Cadence Virtuoso

3. THEORY

The Elmore delay is a first-order approximation used to estimate the propagation delay of an RC network. For an RC ladder, the delay at a particular node depends on the resistance along the path to that node and the capacitances downstream of each resistance.

The Elmore delay at a node is calculated using $t_E = \sum_k R_i C_{downstream,i}$, where R_i is the resistance in the path to the node and $C_{downstream,i}$ is the total capacitance connected downstream of that resistance.

The given circuit consists of five resistors and five capacitors forming an RC lumped network, with five points of interest P1 to P5. The transient response is simulated in Cadence Virtuoso, and the time taken by each point to reach 10%, 50%, and 90% of its final voltage is measured.

The simulated delay values are compared with the corresponding Elmore delay calculated for each point to verify the accuracy of the Elmore delay approximation.

4. TABLES

Table 4.1: Design Parameters

Parameter	Value	Unit
V_{dd}	5	V
R1	10	Ω
R2	12	Ω
R3	14	Ω
R4	16	Ω
R5	18	Ω
C1	12	pF
C2	14	pF
C3	16	pF
C4	18	pF
C5	10	pF

Table 4.2: Result comparison between simulation and theoretical values at 50% of V_{DD}

Point	Elmore RC	Calculated Delay(0.7*RC)	Simulated Delay	Error	Unit
P1	700	490	220.58	-54.9837%	ps
P2	868	607.6	441.22	-27.3831%	ps
P3	1316	921.2	891.4	-3.2349%	ps
P4	1604	1122.8	1213.64	+8.0905%	ps
P5	1496	1047.2	1090.99	+4.1816%	ps

Table 4.3: Result comparison between simulation and theoretical values at 10% of V_{DD}

Point	Elmore RC	Calculated Delay(0.1*RC)	Simulated Delay	Error	Unit
P1	700	70	13.9	-80.1429%	ps
P2	868	86.8	90.46	+4.2166%	ps
P3	1316	131.6	130.4	-0.9119%	ps
P4	1604	160.4	332.43	+107.2506%	ps
P5	1496	149.6	274.14	+83.2487%	ps

Table 4.4: Result comparison between simulation and theoretical values at 90% of V_{DD}

Point	Elmore RC	Calculated Delay(2.2*RC)	Simulated Delay	Error	Unit
P1	700	1540	2059.63	+33.7422%	ps
P2	868	1909.6	2239.67	+17.2848%	ps
P3	1316	2895.2	3059.1	+5.6611%	ps
P4	1604	3528.8	3382.81	-4.1371%	ps
P5	1496	3291.2	3252.14	-1.1868%	ps

5. CALCULATION FOR TRANSMISSION GATE BASED D FLIP FLOP

$$C_{total} = C_1 + C_2 + C_3 + C_4 + C_5 = 12 + 14 + 16 + 18 + 10 = 70pF$$

$$t_{E,P1} = R_1(C_1 + C_2 + C_3 + C_4 + C_5)$$

$$= 10(12 + 14 + 16 + 18 + 10) = 10(70) = 700ps$$

$$t_{E,P2} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_2(C_2)$$

$$= 10(70) + 12(14) = 700 + 168 = 868ps$$

$$t_{E,P3} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_3(C_3 + C_4 + C_5)$$

$$= 10(70) + 14(44) = 700 + 616 = 1316ps$$

$$t_{E,P4} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_3(C_3 + C_4 + C_5) + R_4(C_4)$$

$$= 10(70) + 14(44) + 16(18) = 700 + 616 + 288 = 1604ps$$

$$t_{E,P5} = R_1(C_1 + C_2 + C_3 + C_4 + C_5) + R_3(C_3 + C_4 + C_5) + R_5C_5$$

$$= 10(70) + 14(44) + 18(10)$$

$$= 700 + 616 + 180 = 1496ps$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

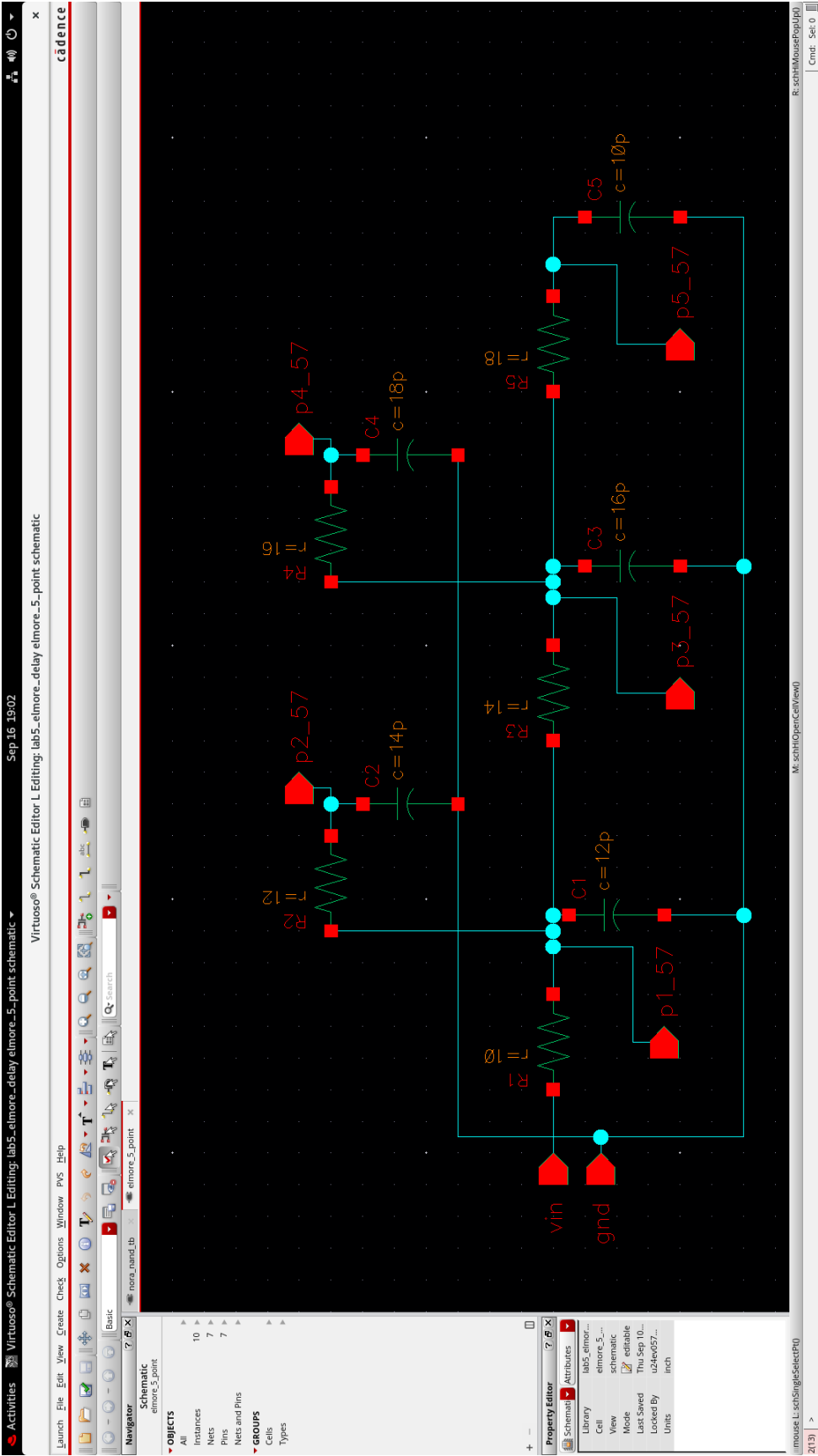


Figure 6.1: Schematic